

OptiCrop: Smart Agricultural Production Optimization Engine

Final Project Report

Project Title

OptiCrop: Smart Agricultural Production Optimization Engine

1. Abstract

The OptiCrop Smart Agricultural Production Optimization Engine is a machine learning–based web application designed to recommend the most suitable crop based on soil nutrients and environmental conditions. The system uses historical agricultural data containing Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall values. Multiple machine learning algorithms were evaluated, and the best-performing model was deployed through a Flask backend API with a responsive web interface.

The project demonstrates the complete lifecycle of a machine learning application, including data preprocessing, exploratory data analysis, model training, evaluation, deployment, and user acceptance testing.

2. Introduction

Agriculture plays a critical role in food security and economic development. Farmers often face challenges in selecting the most appropriate crop for their land due to varying soil and weather conditions. The OptiCrop system addresses this problem by providing data-driven crop recommendations using machine learning techniques.

Objectives

Core goals

- Build a machine learning model for crop recommendation.
- Develop a Flask-based prediction API.
- Create a user-friendly web interface.
- Validate the system through functional and user acceptance testing.

3. Problem Statement

Farmers may not always have access to expert agricultural advice. Incorrect crop selection can lead to reduced yield and financial loss. Therefore, an intelligent recommendation system is required to assist in selecting the optimal crop based on soil and environmental parameters.

4. Scope of the Project

Included

Implemented

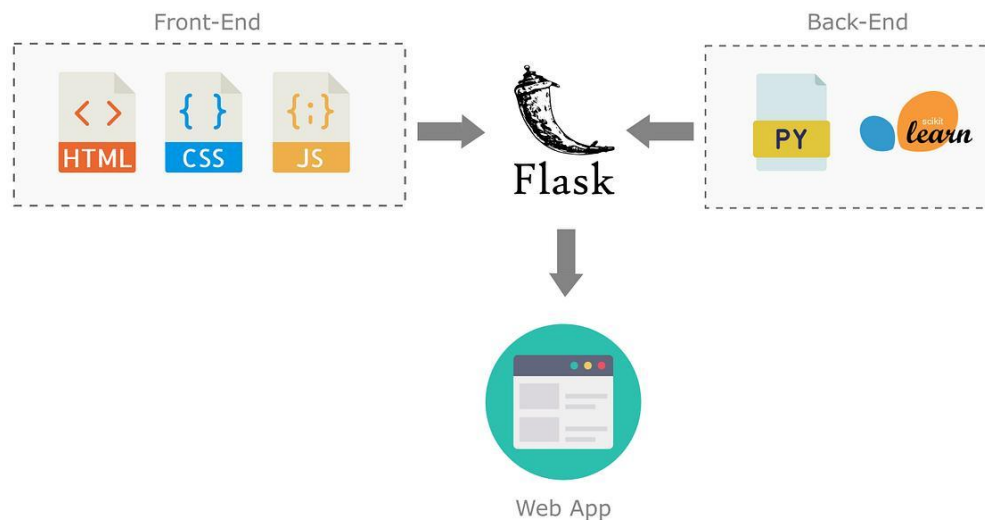
- Soil data analysis
- Machine learning model training
- Crop prediction API
- Web-based dashboard
- Input validation
- Testing and evaluation

Future Scope

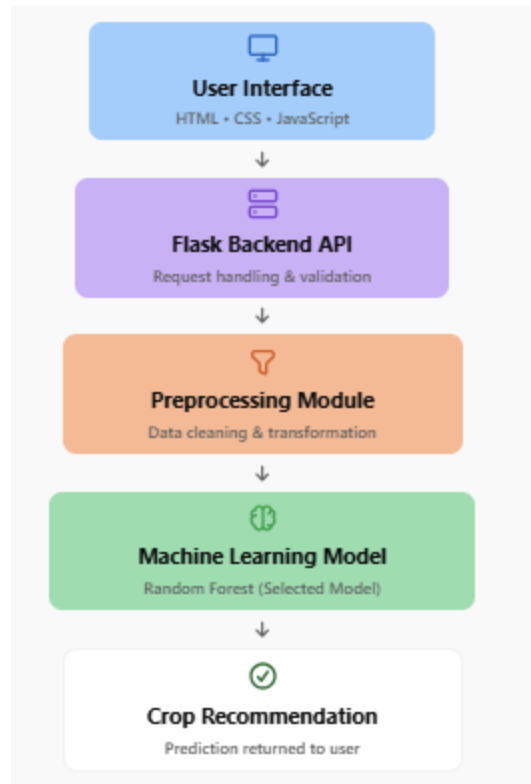
Planned

- Weather API integration
- IoT sensor connectivity
- Cloud deployment
- Mobile application support
- Multilingual interface

5. System Architecture



Architecture Flow



6. Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	Flask (Python)
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Model Storage	Pickle (.pkl)
Development Tools	Jupyter Notebook, VS Code
Dataset	CSV File

7. Machine Learning Implementation

Algorithms Evaluated

4 supervised + 1 unsupervised

Logistic Regression

K-Nearest Neighbors (KNN)

Decision Tree

Random Forest

K-Means Clustering (EDA)

Selected Model

Best performance

Random Forest Classifier

Chosen for highest prediction accuracy and stability

Production model

8. Data Preprocessing

- Handled missing values
- Checked outliers in N, P, and K values
- Normalized numeric ranges
- Analyzed feature correlations
- Prepared training and testing datasets

9. Functional Testing Summary

Functional Test Results

All passed

Metric	Result
Total Test Cases	15
Passed	15
Failed	0
Success Rate	100%

Key Functional Tests

Input validation

Boundary value testing

API request/response testing

Prediction generation

Error handling verification

10. User Acceptance Testing (UAT) Summary

User Acceptance Results

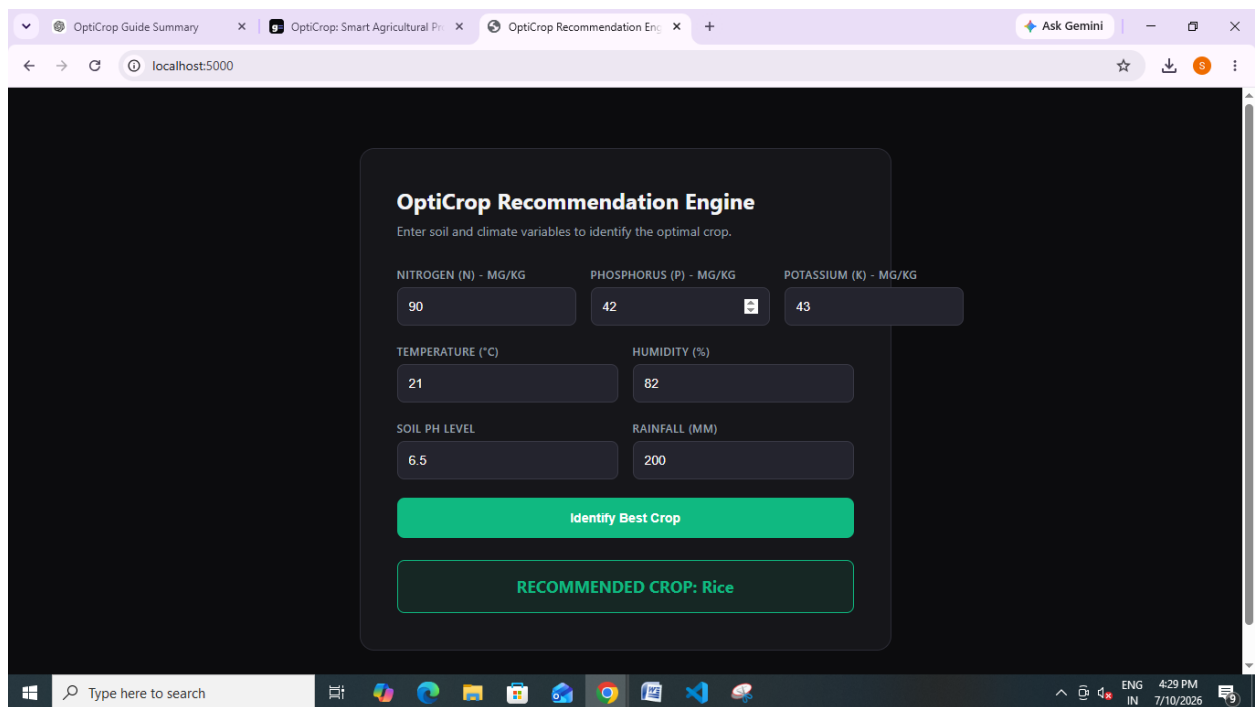
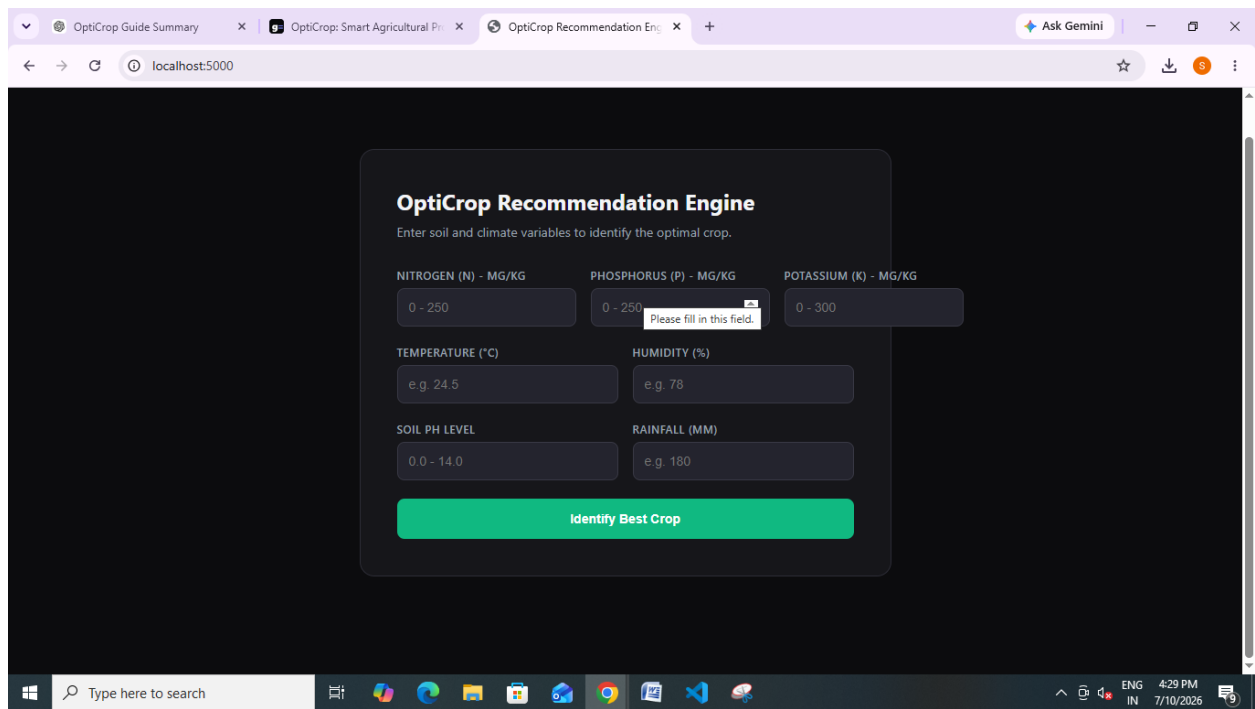
Approved

Metric	Result
Total Test Cases	10
Passed	10
Failed	0
Success Rate	100%

User Feedback

Category	Feedback
Ease of Use	User-friendly interface
Performance	Fast prediction response
Accuracy	Relevant crop recommendations
Navigation	Easy to operate

11. Results and Achievements



Dataset Processed

100%

All records cleaned and prepared

Models

Algorithms Compared

4

Evaluated for performance

Testing

Functional Tests Passed

15 / 15

Zero failures recorded

UAT

User Acceptance Passed

10 / 10

Approved for deployment

12. Future Enhancements

- Integrate live weather forecasting APIs
- Support regional languages
- Deploy on cloud platforms
- Add mobile application support
- Connect with IoT soil sensors
- Provide fertilizer recommendations
- Generate crop yield predictions

13. Conclusion

The OptiCrop Smart Agricultural Production Optimization Engine successfully demonstrates the application of machine learning in agriculture. The system preprocesses soil and environmental data, evaluates multiple algorithms, deploys the best-performing model through a Flask API, and provides real-time crop recommendations through a web interface.

Functional testing and User Acceptance Testing confirmed that the application meets its intended requirements with a 100% test success rate. The modular architecture allows future expansion with weather APIs, cloud deployment, IoT integration, and mobile accessibility, making OptiCrop a scalable and practical solution for intelligent agricultural decision-making.

✓ Project Status

Ready for Submission



Development Completed



Testing Completed



Deployment Ready